Ark Malhotra

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EDUCATION

Course	College / University	Year	CGPA / %
B.Tech (CSE)	Netaji Subhas University of Technology	2028	7.98
Board (Class XII)	Amity International, Saket	2024	93%
Board (Class X)	Amity International, Saket	2022	94.8%

EXPERIENCE

- **Research Intern @ IIT Madras** **Jan 2026 - Jun 2026**
  - Co-developed PyAMorph: 21,000+ LOC **high performance physics Python library** for 2D/3D geometry modelling; built .stl import, GPU acceleration (CuPy + NVIDIA Warp, 320× speedup on geometry eval, 15–100× on mesh-to-SDF), test suite, and reproducible CPU/GPU builds with Pixi
  - Extended AMReX (massively parallel C++ framework) integration with a unified API targeting NumPy, CuPy, and CPU/GPU AMReX grids from the same geometry object.

PROJECTS

- **Backend for FloatChat: AI-powered Conversational Oceanographic Data Visualisation**
  - Built FastAPI + Docker backend for a 20,000+ LOC SIH 2025 submission enabling AI-driven querying of ARGO oceanographic data (5 billion ocean observations, 18,000 NetCDF files) via MCP, with a RAG pipeline (LangChain + PydanticAI + DuckDB); team of 6 ranked top 90 of 200+ college entries
- **Personal Portfolio and Blog Website**
  - Personal portfolio website built using React; implements responsive-design, a markdown engine, and an RSS feed.
- **Hand Cricket Multi-Agent Bandit model**
  - **Python backend** and **React frontend** for an AI (Adversarial Multi-armed-bandit using Experts-As-Arms) to learn from and beat people at the game of playing hand cricket.
  - Composed of **PyTorch neural networks**, n-Gram expert models, and other heuristics woven through the Hedge algorithm.

POSITIONS OF RESPONSIBILITY

- **ML Mentor @ Google Developer Group, NSUT** **Oct 2025 - Present**
  - Designed a Kaggle competition competition with an original dataset, drawing 200+ teams from 4 countries; screened 600+ applicants down to <20 to 6 after interviews
  - Contributed to projects such as makemore (character level text generator using PyTorch), AI factchecking, and the AI backend for Zero Day Apocalypse and BLACKBOX.AI events.
- **Executive Committee @ Axiom, NSUT** **Sep 2025 - Present**
  - Made society's official **website** at axiomnsut.in using **React** with 7 games and newsletter with markdown support

TECHNICAL SKILLS

- **Technical Skills & Tools**: HTML, CSS, JavaScript, TypeScript, Git, Web Development, Machine Learning, Artificial Intelligence, Object Oriented Programming, LLM, Agentic AI
- Ranked **top 17%** in an **international ML Kaggle competition** with a team of three, using libraries such as scikit-learn, Pandas, Numpy, Matplotlib, and Seaborn for EDA and baseline models.